AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 2 · WORKBOOK

Architecture and data flow

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete architecture and data-flow record.

Keep component jobs, trust boundaries, source roles and actual stage traces together.

Keep one habit: Recall → W1–W3/W5 → independent earliest stop → project architecture → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Architecture assigns responsibilities to **component**s and shows information crossing trust boundaries. A diagram should include failure paths.

- Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.

Requirements and service boundaries

> **My remembered idea, example and correction**
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> |---|
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> | |
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> | |

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Architecture route: request validator → trusted-role filter → approved source lookup → drafting → response validator → result. Invalid input stops before lookup. Access filtering must occur before restricted content enters lookup/context; hiding a later citation does not undo exposure.

  Toy sources S student-visible time 16:00 and T teacher-only time 18:00; caller student supplied by trusted test harness, not request body or authentication. Exact request shape {source_id: nonempty string}. Student S path reaches all stages and returns S/16:00. Student T stops denied after role_filter, no source body lookup or draft. Missing X returns review after lookup, no draft.

  Incorrect draft S/19:00 returns review after output check. A request-added role field is invalid, not a promotion. Separate library picture uses approved R/17:30; without retrieval return review. Diagram must be compared with event traces. Deterministic simulation has no model call, network, production authentication or complete security/correctness guarantee.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Architecture route: request validator → trusted-role filter → approved source lookup → drafting → response validator → result. Invalid input stops before lookup. Access filtering must occur before restricted content enters lookup/context; hiding a later citation does not undo exposure.

  Toy sources S student-visible time 16:00 and T teacher-only time 18:00; caller student supplied by trusted test harness, not request body or authentication. Exact request shape {source_id: nonempty string}. Student S path reaches all stages and returns S/16:00. Student T stops denied after role_filter, no source body lookup or draft. Missing X returns review after lookup, no draft.

  Incorrect draft S/19:00 returns review after output check. A request-added role field is invalid, not a promotion. Separate library picture uses approved R/17:30; without retrieval return review. Diagram must be compared with event traces. Deterministic simulation has no model call, network, production authentication or complete security/correctness guarantee.

- A student request bypasses the role filter and sends teacher records directly to a generator. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Architecture route: request validator → trusted-role filter → approved source lookup → drafting → response validator → result. Invalid input stops before lookup. Access filtering must occur before restricted content enters lookup/context; hiding a later citation does not undo exposure.

  Toy sources S student-visible time 16:00 and T teacher-only time 18:00; caller student supplied by trusted test harness, not request body or authentication. Exact request shape {source_id: nonempty string}. Student S path reaches all stages and returns S/16:00. Student T stops denied after role_filter, no source body lookup or draft. Missing X returns review after lookup, no draft.

  Incorrect draft S/19:00 returns review after output check. A request-added role field is invalid, not a promotion. Separate library picture uses approved R/17:30; without retrieval return review. Diagram must be compared with event traces. Deterministic simulation has no model call, network, production authentication or complete security/correctness guarantee.

- A report omits this qualification: A **component** diagram is a plan, not **evidence** that the running implementation enforces its arrows.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Architecture route: request validator → trusted-role filter → approved source lookup → drafting → response validator → result. Invalid input stops before lookup. Access filtering must occur before restricted content enters lookup/context; hiding a later citation does not undo exposure.

  Toy sources S student-visible time 16:00 and T teacher-only time 18:00; caller student supplied by trusted test harness, not request body or authentication. Exact request shape {source_id: nonempty string}. Student S path reaches all stages and returns S/16:00. Student T stops denied after role_filter, no source body lookup or draft. Missing X returns review after lookup, no draft.

  Incorrect draft S/19:00 returns review after output check. A request-added role field is invalid, not a promotion. Separate library picture uses approved R/17:30; without retrieval return review. Diagram must be compared with event traces. Deterministic simulation has no model call, network, production authentication or complete security/correctness guarantee.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Input** is malformed, yet lookup runs and then a validator rejects **output**. Locate the first stage that should stop.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled architecture and **data flow evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of architecture and **data flow**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Architecture and **data flow**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Architecture and **data flow**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Input** is malformed, yet lookup runs and then a validator rejects **output**. Locate the first stage that should stop.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

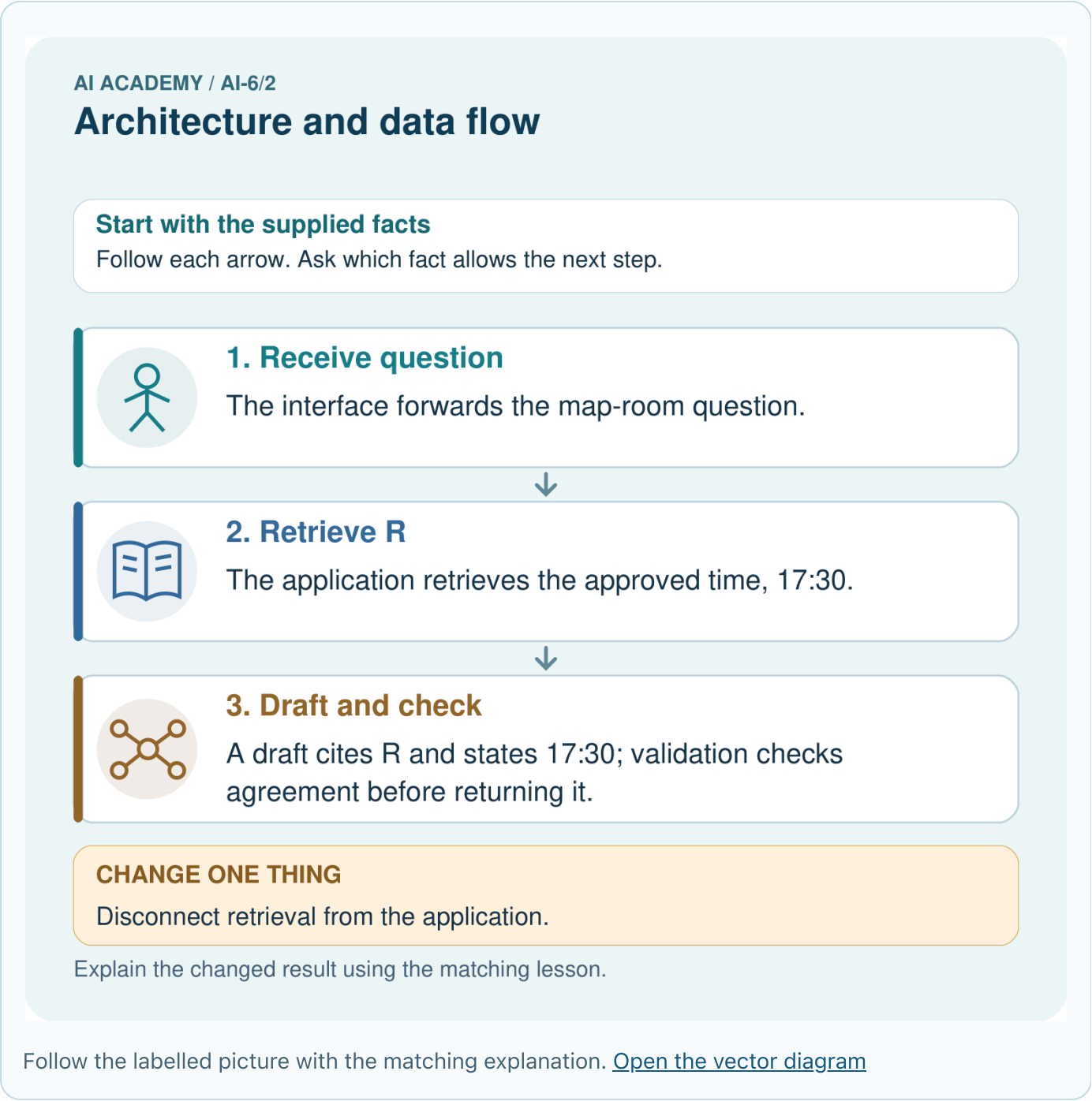
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.
- Approved schedule R says 17:30. The model may phrase facts but cannot replace them.
- Label what moves along the library request, retrieval and drafting arrows. Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service.
- Which labelled step matches this detail? A draft cites R and states 17:30; validation checks agreement before returning it.
- 1. Retrieve R
- 2. Draft and check

- 3. Receive question

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Draft and check A draft cites R and states 17:30; validation checks agreement before returning it.

An architecture assigns responsibilities and carries data between **component**s. Retrieval provides **evidence**; drafting supplies wording.

Changed-case question and optional hint

- Changed-case practice: Disconnect retrieval from the application. Which conclusion is supported by the supplied facts?

- 1. Return review status rather than inventing a time.

- 2. This one example guarantees correct results for every future **input**.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Return review status rather than inventing a time. The supported-answer path lacks its required **evidence input**.

Repair and retain the record

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

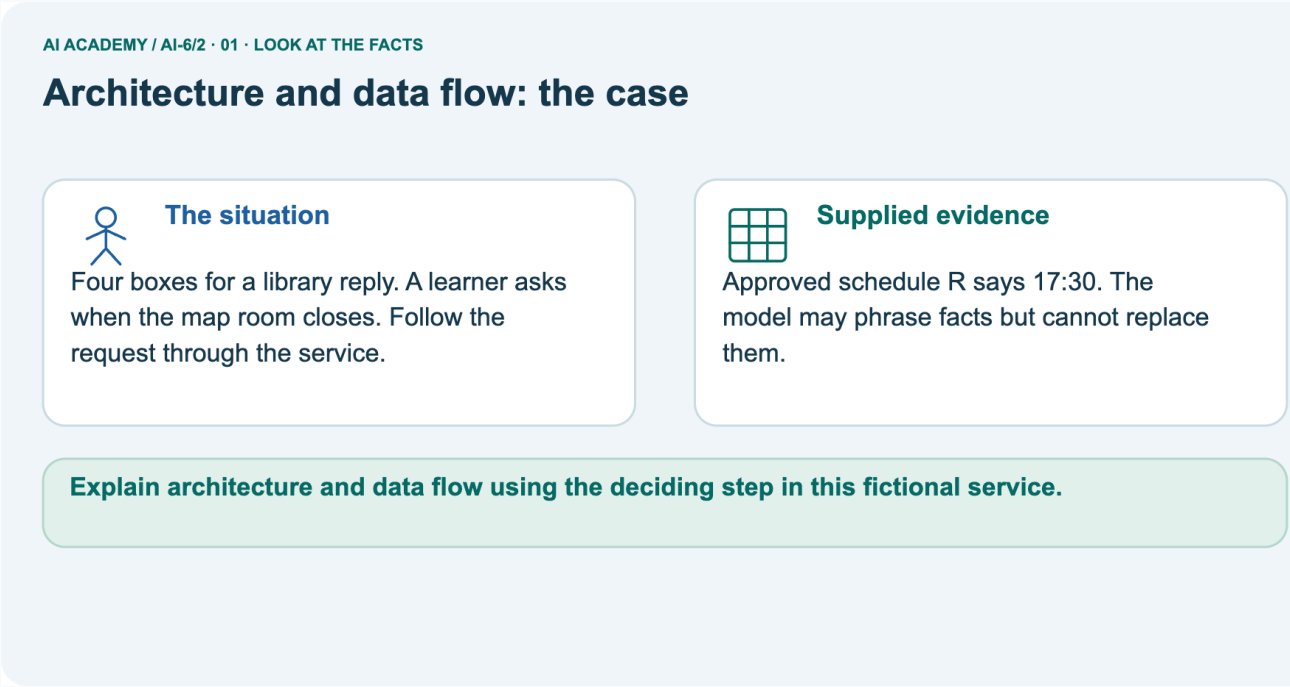
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Architecture and data flow: the case



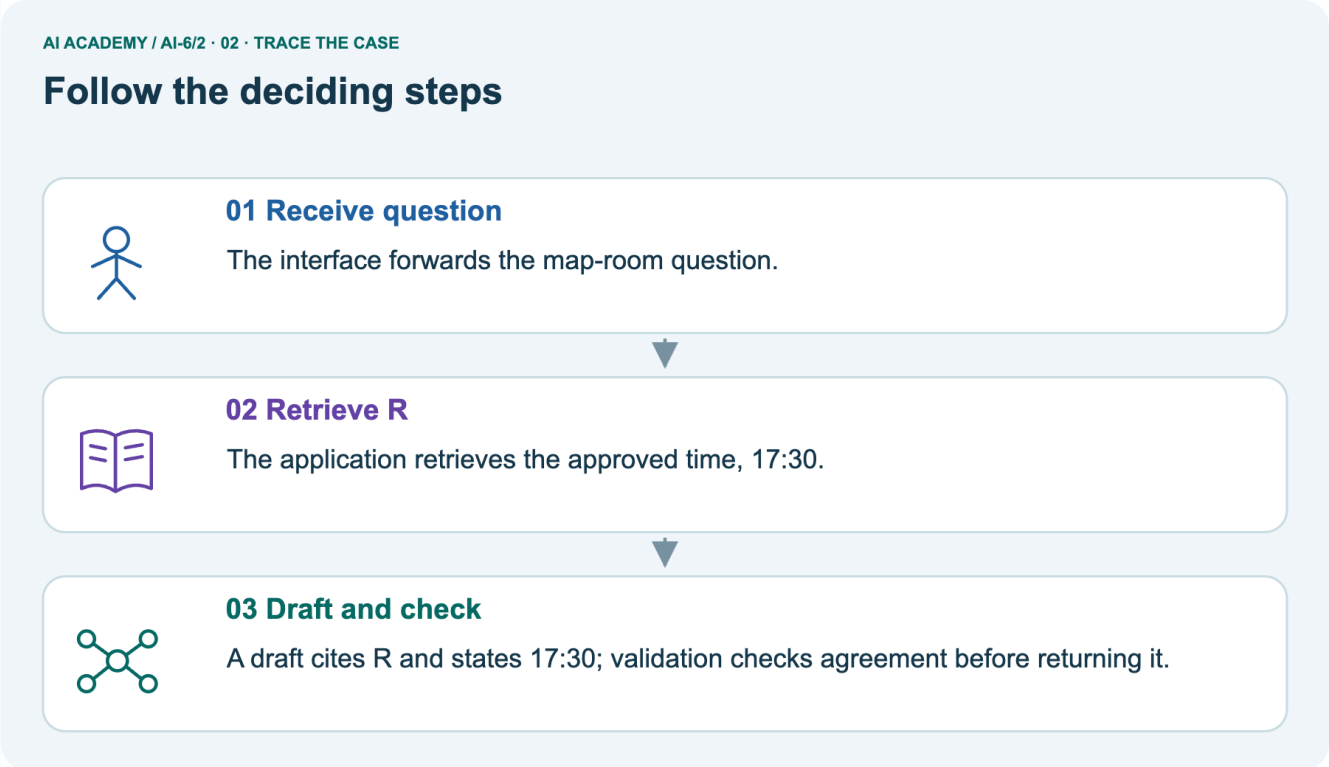
Follow the labelled picture with the matching explanation. Open the vector diagram

Four boxes for a library reply. A learner asks when the map room closes. Follow the request through the service. Supplied facts: Approved schedule R says 17:30. The model may phrase facts but cannot replace them.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



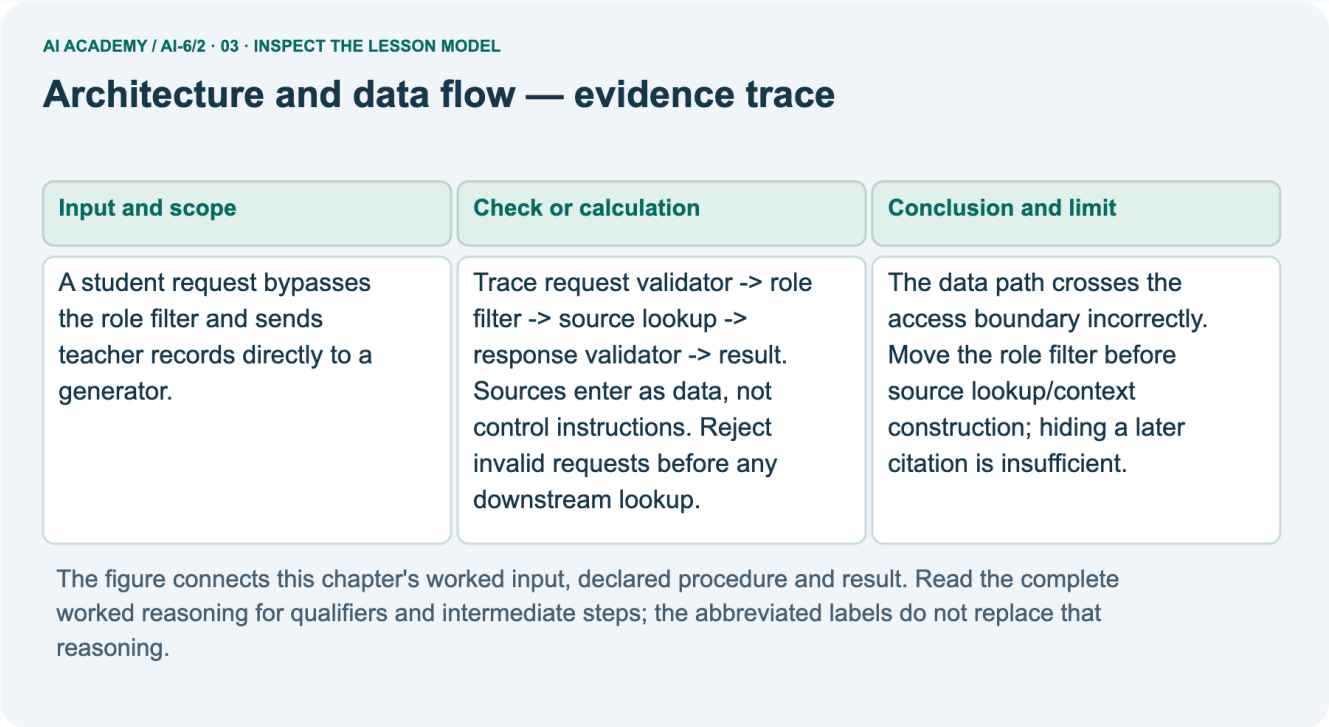
Follow the labelled picture with the matching explanation. Open the vector diagram

Receive question: The interface forwards the map-room question. → Retrieve R: The application retrieves the approved time, 17:30. → Draft and check: A draft cites R and states 17:30; validation checks agreement before returning it.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Architecture and data flow — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Architecture and **data flow** — **evidence** trace.

A student request bypasses the role filter and sends teacher records directly to a generator. | Trace request validator -> role filter -> source lookup -> response validator -> result.

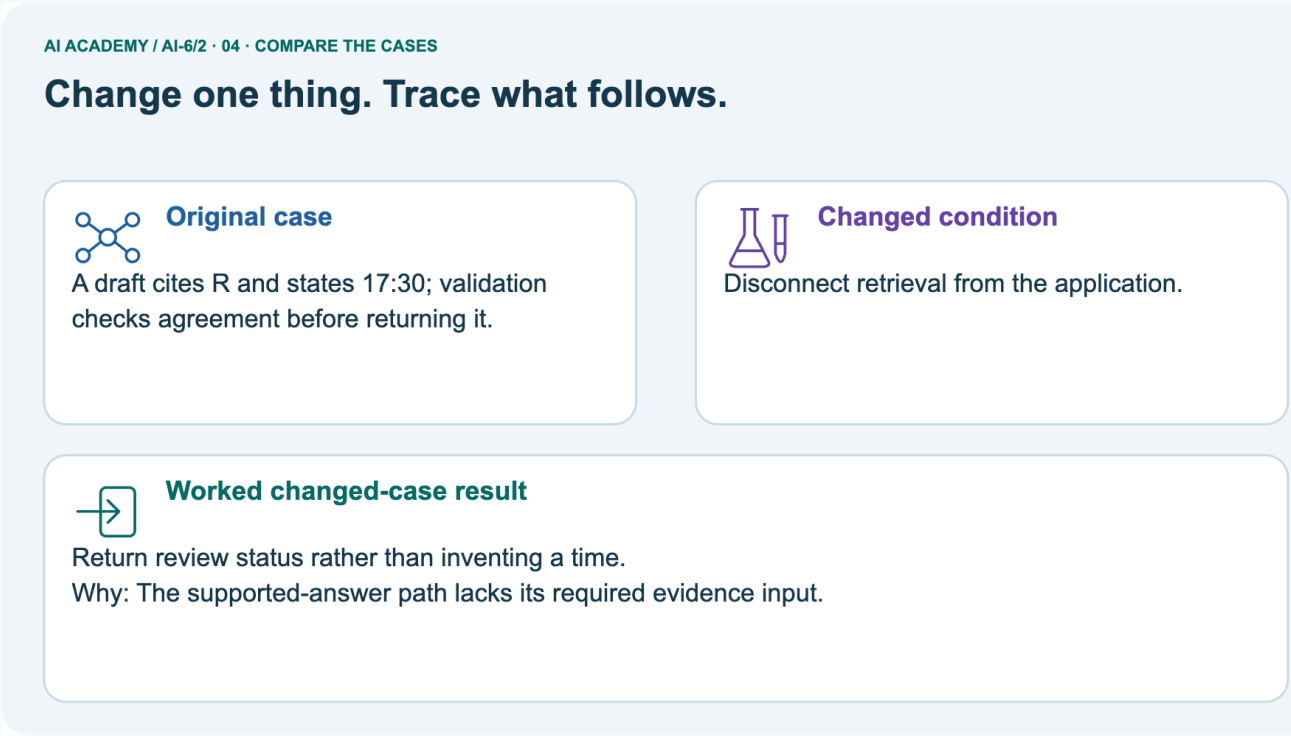
Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup. | The data path crosses the access boundary incorrectly.

Move the role filter before source lookup/context construction; hiding a later citation is insufficient.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

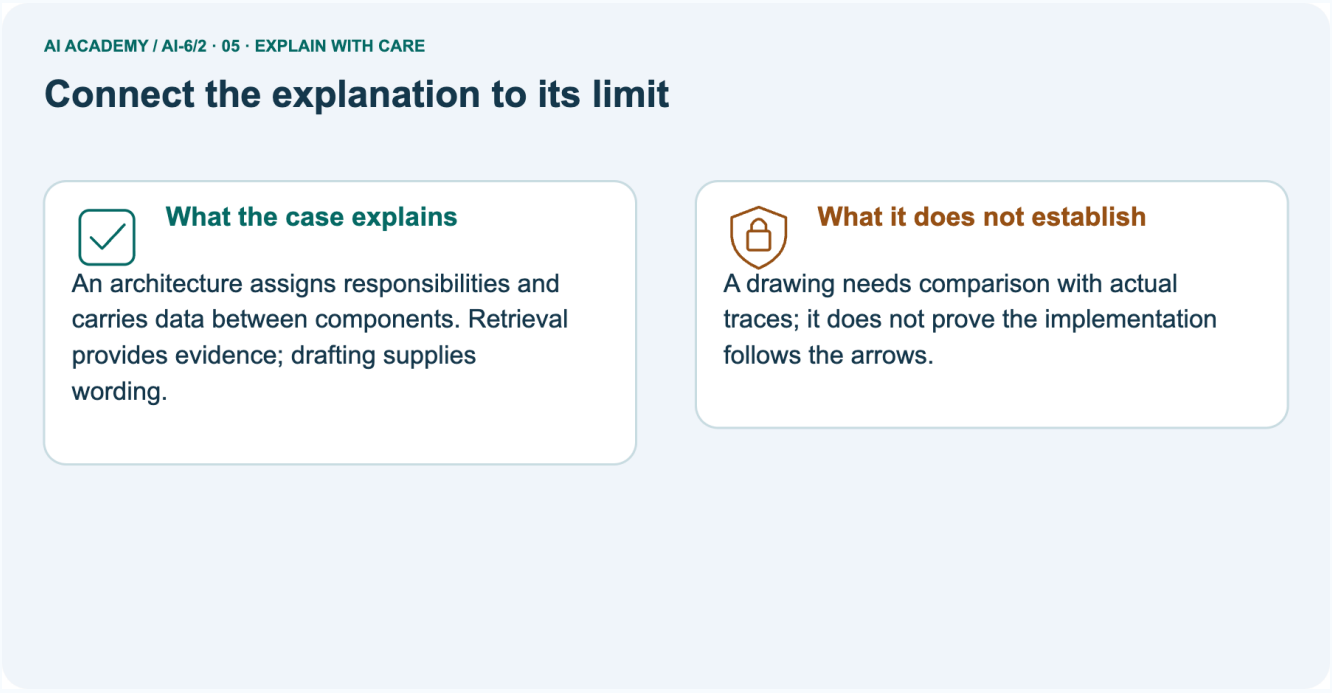
Original case: A draft cites R and states 17:30; validation checks agreement before returning it. Changed condition: Disconnect retrieval from the application.

Expected result: Return review status rather than inventing a time. Reason: The supported-answer path lacks its required **evidence input**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

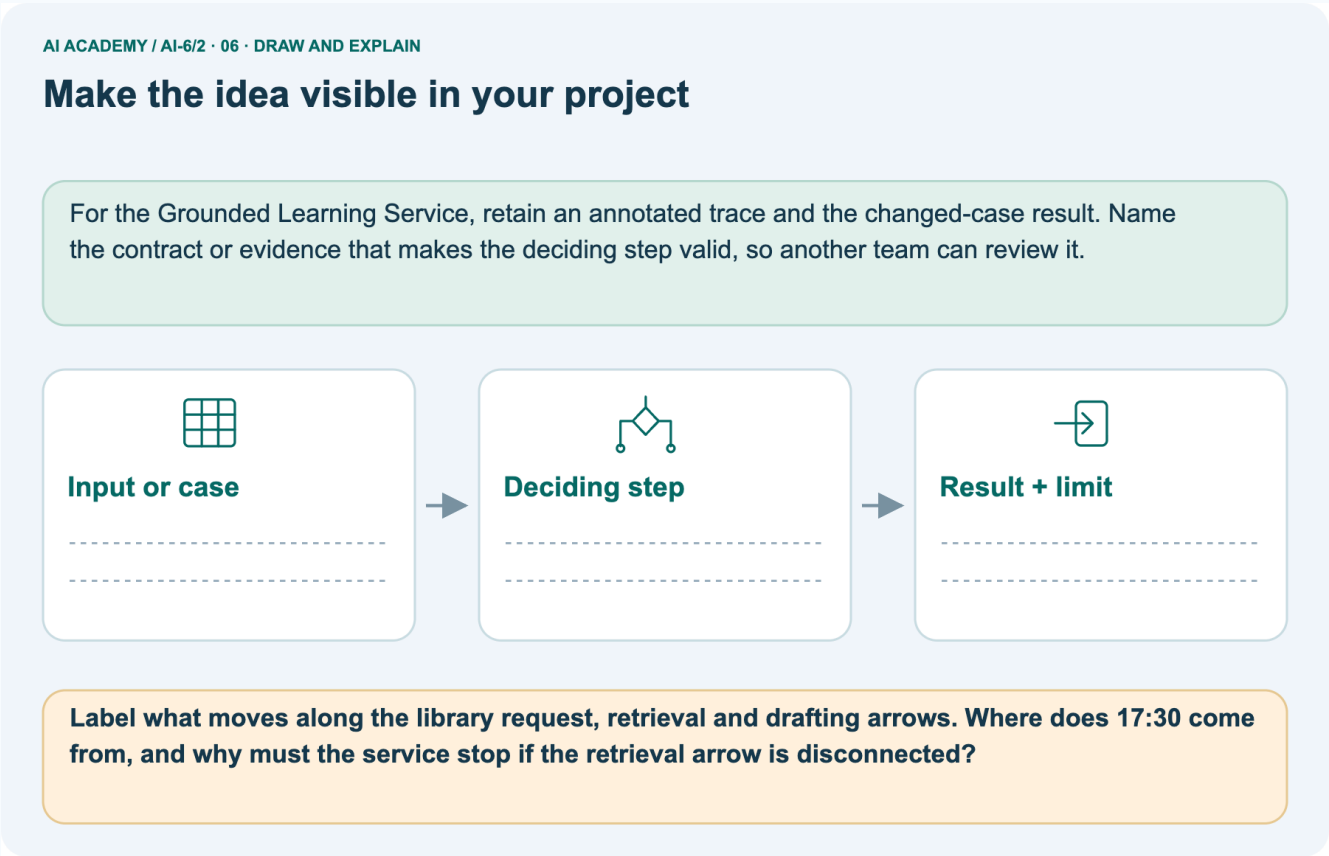
An architecture assigns responsibilities and carries data between **component**s. Retrieval provides **evidence**; drafting supplies wording.

Limit: A drawing needs comparison with actual traces; it does not prove the implementation follows the arrows.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ **Earlier wording, alternatives and picture transcripts**

01. Architecture and data flow: the case

02. Follow the deciding steps

03. Architecture and data flow — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.